Petey HW3 1.

1

$$M(a+bX) = \frac{1}{N}\sum_{i=1}^{N}(a+bX_i) = \frac{1}{N}b\sum_{i=1}^{N}X_i + \cancel{N}a$$

$$\frac{b}{N}\sum_{i=1}^{N}X_i + a = b\,M(X) + a \checkmark$$

$$2.\quad \mathrm{Cov}(X, a+bY) = \frac{1}{N}\sum_{i=1}^{N}(X_i - m(X))(Y_i - m(a+bY))$$

$$X_i - \frac{1}{N}\sum_{i=1}^{N}X_i \qquad Y_i - \frac{1}{N}\sum_{i=1}^{N}(a+bY)$$

$$Y_i = Y_i - a + bY$$

Section

$$\mathrm{Cov}(X, a+bY) = \frac{1}{N}\sum_{i=1}^{N}(X_i - m(X))(a+by_i - m(a+bY))$$

$$a+b\,m(Y)$$

$$a+by_i - a - b \times m(Y) = by_i - b\,m(Y) = \underbrace{b(Y_i - m(Y))}_{\text{single term}}$$

$$\underline{\mathrm{Cov}(X, a+bY) = (X_i - m(X))(Y_i - m(Y)) \cdot B}$$

3. cov $a+bX = b\times cov(X,Y)$

Because there is 2 now,

$(\cancel{a}+bx_i) - (\cancel{a}+b\,m(X))\ \ (\cancel{a}+bx_i) - \cancel{(a}+b\,m(X))$

$$b(x_i - m(X)),\ b(x_i - m(X)) = b^2 (x_i - M(X))(x_i - m(X))$$
$$= b^2 s^2 \text{ or if not given } b^2\text{, then } cov(x,x) = s^2$$

4. Yes because the order would stay the same as long as it's not decreasing, therefore it would be the same place just transformed, and IQR if it's just a shift, range is the same, otherwise min/max kept. Quantile of points stay the same, just not values.

5. No because variables could change in a way swapping signs and values, like log or ~~othe~~ sin, this would create differences in listed values.